Integrated Degree in Psychology

Foundation Year 2024-25

PS50008A: Research Methods

EXPERIMENTAL DESIGN AND STATISTICS - PRACTICE ASSESSMENT

You will need a statistics textbook/handouts/slides and a calculator.

A researcher decided to investigate whether pairs of friends are similarly extravert by giving Eysenck Personality Inventories (E.P.I.) to volunteers and their best friends at a university.

The E.P.I. produces a score for extraversion (max=25, where a low score=introversion and a high score=extraversion) and a score for neuroticism (max=25, where a low score=emotional stability and a high score=neuroticism), and according to Eysenck's theory, the two personality dimensions are defined as entirely independent of one another, i.e. the extraversion score is quite separate from/unrelated to the neuroticism score.

The extraversion data were as follows:

Participant Pair	Person 1	Person 2
1	10	12
2	7	13
3	14	17
4	11	5
5	13	9
6	12	14
7	6	8
8	18	12
9	21	15
10	9	11

Answer the following questions and include ALL your calculations:

1) State the researcher's hypothesis.

2) Calculate the means and standard deviations for each set of scores.

3) Conduct an analysis to see if there is a significant correlation between pairs of friends' extraversion scores.

4) If there was no significant correlation, can you think of two reasons why not. (You can suggest methodological and/or theoretical reasons.)

5) Would you expect there to be a significant correlation between pairs of friends' neuroticism scores – and state two reasons why. (These should really be different from those given for Qu. 4 and again, can be methodological and/or theoretical.)